SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I M.Dhanushkumar (192511429) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name: M. Dhanush kumar

Reg no: 192511429

Course Code: CSA

Course Name: Cryptography and Network security

1- Event communication security (caesar cipher)

a) Reconstruct encrypted message if a shift of 4 position is applied.

Given:

MEET AT NOON

Step by step encryption using a shift 4 letters

M → Q
E → I
E → I
T → X
A → E
T → X
N → R
O → S
O → S
N → R

Answer: QIIX EX RSSR

∴ Spaces remain unchanged

b) Explain how an authorized member can retrieve original message:

The authorized member decrypts the message by shifting every letter 4 position backward in the alphabet.

Ex: Q → M
I → E
I → E
X → T

∴ MEET AT NOON → QIIX EX RSSR

2- Intercepted suspicious message (Substitution cipher)

a) Deduce the most likely original message

Given:

Encrypted message:

GSRH RH Z HVXIVG

using the Atbash cipher, where, A<->Z, B<->Y, Y, C<->X

G -> T
S -> H
R -> I
H -> S
R -> I
H -> S
Z -> A
H -> S
V -> E
X -> C
I -> R
V -> E
G -> T

Answer: THIS IS A SECRET

b) Justify your answer using letter patterns or cipher characteristics:

The encrypted text contains the same number of letters as the original message

The Atbash cipher replaces each letter with its opposite in the alphabet.

Applying the same substitution rule to the ciphertext produces meaningful English words.

∴ Original message is "THIS IS A SECRET

2. Secure Messaging system (Vigenere Cipher).

a) Demonstrate how encryption transforms the message "HELLO" using the "key"

Given:-

Plaintext : HELLO

keyword : KEY

Repeat the Keyword until it matches the length of the message.

Plainterxt : H E L L O

Repeated key : K E Y K E

Encrypted letter : R I J V S

using the Vigenere cipher, each plaintext letter is shifted forward according to the corresponding keyword letter.

Ex: H + K = R

E + E = I

L + Y = J

L + K = V

O + E = S

Answer:

The encrypted message is "RIJVS"

b) Explain how the repeating keyword influence each character of ciphertext

The keyword "KEY" is repeated as "KEYKE"
Each keyword letter determines the shift applied to its corresponding plaintext letter.
Different keyword letters produce different shifts.

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4. Suspicious Digital Assest investigation (image steganography).
a) Analyze how investigators could detect hidden data within an image.

Metadata analysis:
Examine image properties and metadata for unusual information.

File size analysis
compare the image size with similar images to identify unexpected differences.

statistical analysis:
Analyze pixel values and color distributing for suspicious patterns

steganalysis tools:
use specialized softwcere to identify possible hidden message or embedded files.

Step :1

Arrange the letter of the ciphertext back into the three-rail zigzag pattern

Step :2

Read the letter along the reconstructed zigzag path to recover the original message.

Answer :

WE ARE DISCOVERED FLEE AT ONCE

∴ without space.

WEAREDISCOVEREDFLEEATONCE

b) Critically evaluate why this method is the vulnerable in communication system

1. Small key space :

The number of possible rail arrangement is limited

2. Pattern preservation

The cipher rearrange letter without changing them

3. Frequency analysis :

Letter frequencies remain the same as in the original message.

File comparison:

compare the suspicious image with the original image, if available, to detect changes.

b) Recommend two techniques to strengths the security of hidden communication

1. Encrypt the message before embedding

Encrypt the confidential information using a strong, modern encryption algorithm before hiding it inside the image.

2. use key-based adaptive steganography

Embedded the encrypted data in carefully selected image location using a secret key. The image and make unauthorized extraction more difficult.

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5. Legacy. Encryption in corporate communication (transposition cipher)

a) the most probable original message by the reversing the encryption process.

Given:

Ciphertext

WEARLTEERSOEFEAOCAIVDEN

The message can be decoded using three Rail fence cipher

5. Lack of modern security:

It does not provide strong confidentiality against modern cryptanalysis.

c) Propose one practical enhancement to improve message confidentiality

Use a modern unauthorized encryption algorithm such as AES-GCM with a securely managed key.

AES-GCM encrypts the message and also provides an integrity check to help detect unauthorized modification.